

Medical Diagnosis Expert System Using Prolog

1. Title

Medical Diagnosis Expert System Using Prolog.

2. Aim

To develop a simple rule-based Medical Diagnosis Expert System using Prolog that identifies possible medical conditions based on patient symptoms and provides general advice

3. Objectives

- To understand the basic concepts of Prolog.
- To represent patient symptoms using facts.
- To create diagnosis rules using logical reasoning.
- To identify possible medical conditions.
- To demonstrate the use of Prolog in Artificial Intelligence and Expert Systems.

4. Prolog Program

% Patient Information

patient(tanu).

patient(rahul).

patient(riya).

patient(amt).

% Symptoms

symptom(tanu, fever).

symptom(tanu, cough).

symptom(tanu, sore_throat).

syptom(rahul, headache).

symptom(rahul, nausea).

symptom(rahul, sensitivity_to_light).

symptom(riya, sneezing).

symptom(riya, runny_nose).

symptom(riya, itchy_eyes).

symptom(amit, stomach_pain).

symptom(amit, vomiting).

symptom(amit, diarrhea).

% Diagnosis Rules

has_flu(Patient) :-

 symptom(Patient, fever),

 symptom(Patient, cough),

 symptom(Patient, sore_throat).

has_migraine(Patient) :-

 symptom(Patient, headache),

 symptom(Patient, nausea),

 symptom(Patient, sensitivity_to_light).

has_allergy(Patient) :-

 symptom(Patient, sneezing),

 symptom(Patient, runny_nose),

 symptom(Patient, itchy_eyes).

has_food_poisoning(Patient) :-

 symptom(Patient, stomach_pain),

```
symptom(Patient, vomiting),  
symptom(Patient, diarrhea).
```

```
% Diagnosi
```

```
diagnosis(Patient, flu) :-  
    has_flu(Patient).
```

```
diagnosis(Patient, migraine) :-  
    has_migraine(Patient).
```

```
diagnosis(Patient, allergy) :-  
    has_allergy(Patient).
```

```
diagnosis(Patient, food_poisoning) :-  
    has_food_poisoning(Patient).
```

```
% Advice
```

```
advice(flu, 'Take adequate rest and stay hydrated.').
```

```
advice(migraine,  
    'Rest in a quiet environment and stay hydrated.').
```

```
advice(allergy,  
    'Avoid known allergens and keep the surroundings clean.').
```

```
advice(food_poisoning,  
    'Drink fluids and seek medical attention if symptoms are severe.').
```

```
% Medical Report
```

medical_report(Patient, Disease, Advice) :-

diagnosis(Patient, Disease),

advice(Disease, Advice).

5. Flowchart

START



Store Patient & Symptom Facts



Create Knowledge Base

(Facts + Rules)



Enter Query



Match Symptoms with Rules



Rule Satisfied?

YES NO



Identify No Matching

Condition Diagnosis



Retrieve ↓

Advice ↓



Display Diagnosis & Advice



END

6. Result Analysis

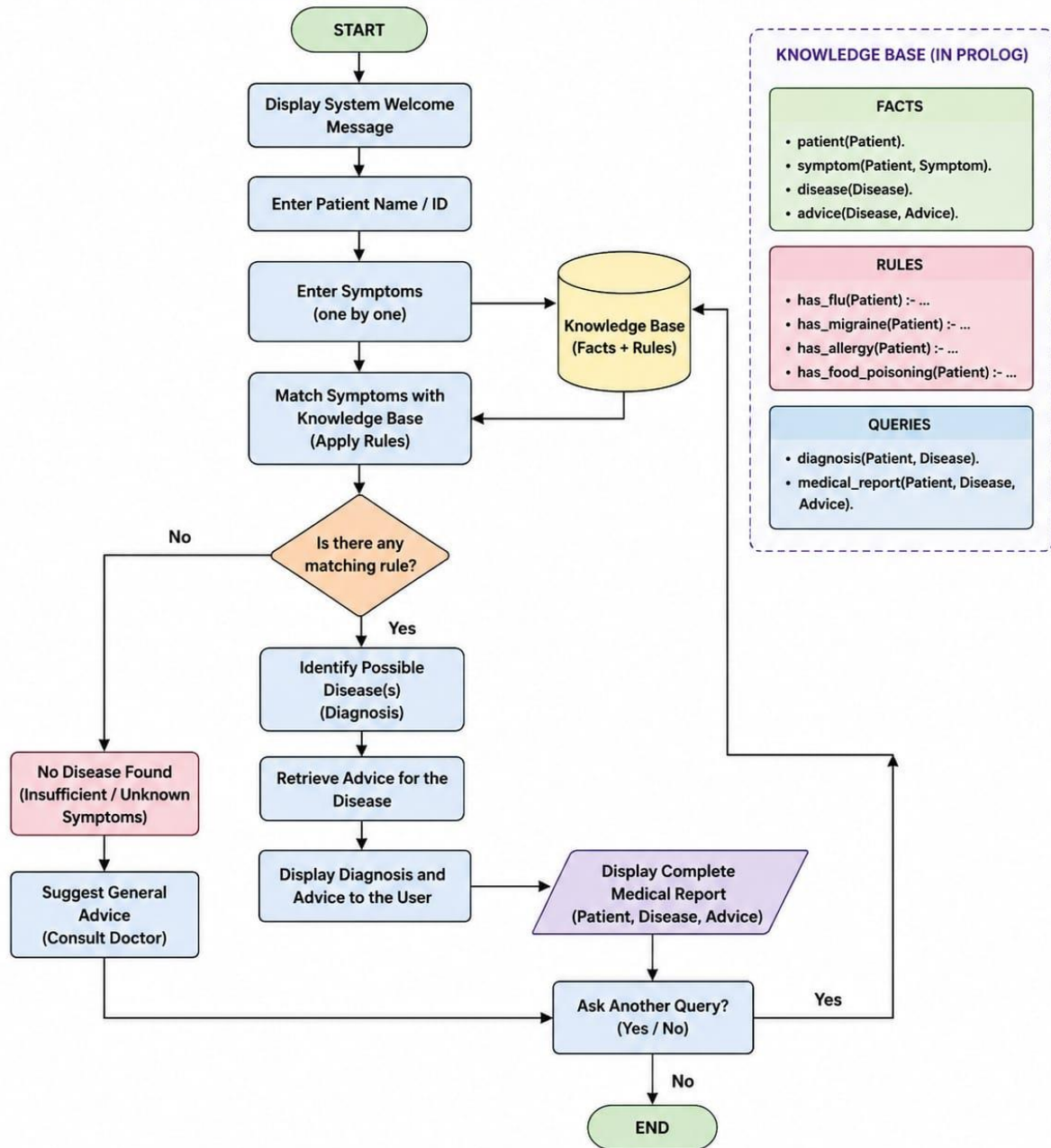
The Medical Diagnosis Expert System successfully identifies possible medical conditions by matching patient symptoms with predefined rules. The system correctly detects conditions such as flu, migraine, allergy, and food poisoning based on the symptoms stored in the knowledge base.

The output demonstrates that Prolog can effectively perform logical reasoning using facts and rules. The system also provides suitable general advice related to the identified condition.

7. Conclusion

The Medical Diagnosis Expert System demonstrates the use of Prolog for building a simple rule-based expert system. It uses facts, rules, and logical inference to identify possible medical conditions and provide advice. This project illustrates the application of Artificial Intelligence concepts using Prolog.

MEDICAL DIAGNOSIS EXPERT SYSTEM (PROLOG) FLOWCHART



SWISH -- SWI-Prolog for SHarir | Sign in to GitHub · GitHub | Ask Gemini

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Program

```

1 patient(tanu).
2 patient(rahul).
3 patient(riya).
4 patient(amit).
5
6 symptom(tanu, fever).
7 symptom(tanu, cough).
8 symptom(tanu, sore_throat).
9
10 symptom(rahul, headache).
11 symptom(rahul, nausea).
12 symptom(rahul, sensitivity_to_light).
13
14 symptom(riya, sneezing).
15 symptom(riya, runny_nose).
16 symptom(riya, itchy_eyes).
17
18 symptom(amit, stomach_pain).
19 symptom(amit, vomiting).
20 symptom(amit, diarrhea).
21
22 disease(flu).
23 disease(migraine).
24 disease(allergy).
25 disease(food_poisoning).
26
27

```

diagnosis(tanu,Disease)

Disease = flu
false

diagnosis(rahul,Disease)

Disease = migraine
Next 10 100 1,000 Stop

medical_report(tanu,Disease,Advice)

Advice = 'Take adequate rest and stay hydrated.'
Disease = flu
Next 10 100 1,000 Stop

has_flu(tanu)

true

has_food_poisoning(Patient)

Patient = amit

?-
?

Examples History Solutions

table results Run!

23°C Mostly cloudy

21:26 14-08-2026

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Program

```

34 has_migraine(Patient) :-
35     symptom(Patient, headache),
36     symptom(Patient, nausea),
37     symptom(Patient, sensitivity_to_light).
38
39 has_allergy(Patient) :-
40     symptom(Patient, sneezing),
41     symptom(Patient, runny_nose),
42     symptom(Patient, itchy_eyes).
43
44 has_food_poisoning(Patient) :-
45     symptom(Patient, stomach_pain),
46     symptom(Patient, vomiting),
47     symptom(Patient, diarrhea).
48
49
50
51 diagnosis(Patient, flu) :-
52     has_flu(Patient).
53
54 diagnosis(Patient, migraine) :-
55     has_migraine(Patient).
56
57 diagnosis(Patient, allergy) :-
58     has_allergy(Patient).
59
60

```

diagnosis(tanu,Disease)

Disease = flu
false

diagnosis(rahul,Disease)

Disease = migraine
Next 10 100 1,000 Stop

medical_report(tanu,Disease,Advice)

Advice = 'Take adequate rest and stay hydrated.'
Disease = flu
Next 10 100 1,000 Stop

has_flu(tanu)

true

has_food_poisoning(Patient)

Patient = amit

?-
?

Examples History Solutions

table results Run!

21:26 14-08-2026